Result 1 (intersection then union vanishes). [Singular homology inclusion maps, functoriality and signed cancellation]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let U and V be subsets of X. Then

$$\beta_{U,V} \circ \alpha_{U,V} = 0.$$

Here $\alpha_{U,V} = (i_*, -j_*)$ maps intersection homology into $\mathcal{H}(U) \oplus \mathcal{H}(V)$, while $\beta_{U,V} = [k_*, \ell_*]$ maps this sum into union homology. All four maps come from subset inclusions.

This composition-to-zero statement holds for arbitrary subsets. It does not assert exactness or supply a connecting map. The cover and excision conditions needed for Mayer–Vietoris remain separate obligations.

LLM-assisted definitions and supporting proof (gpt-6, 2026-11-10).